

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 313 — Past Questions & Materials
by Renance

Course: PHY 313
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 7
Levels: 300

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Paper 1 of 7 — PHY 313 - 2019/2020 - First Semester - Analytical Mechanics I

Level 300 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria.
College of Science
Department of Physics
First Semester Examination

Academic Session: 2019/2020
Course Title: Analytical Mechanics I
Course Code: PHY313 (3 units)
Time Allowed: 3hrs.

27/02/2021

Instructions: Answer Question 1 and Any other Three Questions. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

Question 1

- A. Prove that the total angular momentum of a system of particles, referred to a given origin, can be expressed as the sum of two parts: one part being the angular momentum of all the particles relative to the centre-of-mass, and the other being the angular momentum of the total mass assumed to be at the centre-of-mass, relative to the origin.
- B. A system of particles consist of four particles of masses one quarter ($\frac{1}{4}$) unit each, with position vectors $\vec{r}_1 = 4t\hat{a}_x - 8\hat{a}_y - 4t^2\hat{a}_z$, $\vec{r}_2 = 12t\hat{a}_x + 4t\hat{a}_y$, $\vec{r}_3 = \vec{r}_1$, $\vec{r}_4 = \vec{r}_2$, where t is the time. Calculate the total angular momentum of the system.

(22 marks)

Question 2

- A. (i) State the Divergence theorem in Cartesian coordinates, and define all terms.
(ii) Given a force $\vec{F} = x^2\hat{a}_x + xy\hat{a}_y + yz\hat{a}_z$, in Cartesian coordinate. Verify the divergence theorem over a cube half ($1/2$) unit on each side. The cube is located at the first octant of the Cartesian system with one corner at the origin.
- B. State the curl of a vector field in Cartesian coordinates, and explain the physical meaning of the expression. (ii) Show that $\text{curl } \vec{B} = 0$, if $\vec{B} = (k/r)\hat{a}_r$, in spherical coordinates, where k is a constant.

(16 marks)

Question 3

- A. Consider a particle moving in a spherical coordinate system. Write down the coordinate transformation equations and obtain the position vector, $\vec{r}$ in spherical coordinate.
- B. Derive an expression for the kinetic energy and potential energy possess by the particle.
- C. Hence, obtain the Lagrangian formulations and derive the Larange's equations for the three generalized coordinates.

(16 marks)

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Question 4

A. Let (r, θ) represents the plane polar coordinates describing the position vector of a particle P. If $\hat{r}$ is a unit vector in the direction of the position vector $\vec{r}$, and $\hat{\theta}$ is a unit vector perpendicular to $\hat{r}$ and in the direction of increasing θ .

- Derive an expression for the velocity of the particle.
- Derive an expression for the acceleration of the particle.
- Show that the differential equation of the path obeyed by the particle of mass m , in terms of u and θ is $f(\pm) = -m(hu)^2 \left(u + \frac{d^2u}{d\theta^2} \right)$.

B. Assuming a particle is under the influence of a central force field $f(r) = \left(\frac{\alpha}{r^2} + \frac{\beta}{r^3} \right)$, where $\alpha, \beta > 0$. Find the orbit or path of the particle.

(16 marks)

Question 5

- Define central force motion and state three characteristics.
- State Kepler's laws of planetary motion, and prove the 3rd law.
- (i) Show that the magnitude of the velocity of a particle at any point on its path as it orbits elliptically the sun is $v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$.

Define the terms v, μ, a, r as it appears in C(i) above.

(16 marks)

Question 6

- Write down the point and vector transformations from (i) Cartesian to Spherical coordinates (ii) Spherical to Cartesian coordinates.
- (i) Given that vector $\vec{A} = A_r \hat{a}_r + A_\theta \hat{a}_\theta + A_z \hat{a}_z$, derive the transformation equation from Cylindrical to Cartesian coordinates in matrix form.

(16 marks)

$$\frac{1}{1} \propto \frac{1}{2} = \frac{1}{2}$$

$$\frac{1}{2}$$

Paper 2 of 7 — PHY 313 - 2017/2018 - First Semester - Analytical Mechanics I

Level 300 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria.

College of Science

Department of Physics

First Semester Examination

Academic Session: 2017/2018

Course Title: Analytical Mechanics I

Course Code: PHY313 (3 units)

Time Allowed: 3hrs. 29/05/2018

Instructions: Answer Question 1 and Any other Three Questions. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

Question 1

Prove that the total angular momentum of a system of particles, referred to a given origin, can be expressed as the sum of two parts: one part being the angular momentum of all the particles relative to the centre-of-mass, and the other being the angular momentum of the total mass assumed to be at the centre-of-mass, relative to the origin.

A system of particles consists of four particles of masses one quarter (+) unit each, with position vectors $\mathbf{r}_1 = 4\mathbf{i} - 8\mathbf{j} - 40\mathbf{k}$, $\mathbf{r}_2 = 12\mathbf{i} + 4\mathbf{j}$, $\mathbf{r}_3 = \mathbf{i}$, $\mathbf{r}_4 = 7\mathbf{j}$, where t is the time. Calculate the total angular momentum of the system.

(22 marks)

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Question 2

As) State the Divergence theorem in Cartesian coordinates, and define all terms.

AMli) Given a force $\mathbf{F} = x^4\mathbf{i} + xy\mathbf{j} + yz\mathbf{k}$, in Cartesian coordinate. Verify the divergence theorem over a cube half (1/2) unit on each side. The cube is located at the first octant of the Cartesian system with one corner at the origin.

BC State the curl of a vector field in Cartesian coordinates, and explain the physical meaning of the expression- $\nabla \times \mathbf{B}$. Show that $\nabla \times \mathbf{B} = 0$, if $\mathbf{B} = (k/r)\mathbf{a}$, in spherical coordinates, where k is a constant.

(16 marks)

Question 3

A. Consider a particle moving in a spherical coordinate system. Write down the coordinate transformation equations and obtain the position vector, $\mathbf{r}$ in spherical coordinate,

B. Derive an expression for the kinetic energy and potential energy possessed by the particle.

C. Hence, obtain the Lagrangian formulations and derive the Lagrange's equations for the three generalized coordinates.

(16 marks)

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Question 4

A. Let $(7, 0)$ represents the plane polar coordinates describing the position vector of a particle P. If $\hat{r}$ is a unit vector in the direction of the position vector $\vec{r}$, and $\hat{\theta}$ is a unit vector perpendicular to $\hat{r}$ and in the direction of increasing θ .

(i) Derive an expression for the velocity of the particle.

(ii) Derive an expression for the acceleration of the particle.

(iii) Show that the differential equation of the path obeyed by the particle of mass m , in terms of w and θ is $f(\theta) = -m(hu)'$ (« oe),

3

B. Assuming a particle is under the influence of a central force field $f(r) = (-k/r^2)$, where $k, \epsilon > 0$. Find the orbit or path of the particle.

(16 marks)

Question 5

. Define central force motion and state three characteristics.

BY State Kepler's laws of planetary motion, and prove the 3rd law.

C. (i) Show that the magnitude of the velocity of a particle at any point of its path as it

orbits elliptically the sun is $v' = u(2-2)$.

ro Gt

(ii) Define the terms v, y, a, r , as it appears in C(i) above,

(16 marks)

Qu rigs 6 : ater

. Write down the point and vector transformations from (i) Cartesian to Spherical coordinates (ii) Spherical to Cartesian coordinates.

BY (i) Given that vector $A = 4, 4, + A, d, + A, a,$, derive the transformation equation from Cylindrical to Cartesian coordinates in matrix form.

(ii) Transform the vector field $A = 4, \sin \theta + 4, \cos \theta + 4, ,$ to Cartesian coordinates.

F a (16 marks)

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Paper 3 of 7 — Phy 313 Test PQ

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
B.Sc (MATHEMATICS) DEGREE SEMESTER EXAMINATION (2019/2020), FIRST SEMESTER.
MTH 313 (Complex Analysis I). February, 2021.
Time: 2hrs.

Instruction: Answer 4 questions
 All Questions Carry Equal Marks

Question 1. (a) Let $z = a + bi$ and $z' = c + di$,
 $u + iv$.
 (b) Verify the Green's theorem.

PHY 313, TEST 1
Answer ALL Questions.

- (a) Write down the mathematical expression of the Divergence Theorem in Cartesian coordinates, and define all terms in the equation.
 (b) Given a force $\vec{F} = x^2\hat{a}_x + xy\hat{a}_y + yz\hat{a}_z$, in Cartesian coordinate. Verify the divergence theorem over a cube two (2) units on each side. The cube is located at the first octant of the Cartesian system with one corner at the origin.
- (a) write down the point and vector transformations from Cartesian to spherical coordinates.
 (b) Write down the point and vector transformations from spherical to Cartesian coordinates.
 (c) Transform the vector field $\vec{A} = \sin\theta\hat{a}_r + \cos\theta\hat{a}_\theta + \hat{a}_\phi$, to Cartesian coordinates.
- (a) Prove that the total angular momentum of a system of particles, referred to a given origin, can be expressed as the sum of two parts: one part being the angular momentum of all the particles relative to the centre-of-mass, and the other being the angular momentum of the total mass assumed to be at the centre-of-mass, relative to the origin.
 (b) A system of particles consist of four particles of masses half ($\frac{1}{2}$) unit each, with position vectors $\vec{r}_1 = 2t\hat{a}_x - 3t\hat{a}_y - t^2\hat{a}_z$, $\vec{r}_2 = t\hat{a}_x + 2t\hat{a}_y + 3t\hat{a}_z$, $\vec{r}_3 = \vec{r}_1$, $\vec{r}_4 = \vec{r}_2$, where t is the time. Calculate the total angular momentum of the system.

(2c) $\vec{A} = \sin\theta\hat{a}_r + \cos\theta\hat{a}_\theta + \hat{a}_\phi$ this is in spherical coord
 thus,

x	$\sin\theta\cos\phi$	$\cos\theta\cos\phi$	$-\sin\phi$	A_x
y	$\sin\theta\sin\phi$	$\cos\theta\sin\phi$	$\cos\phi$	A_y
z	$\cos\theta$	0	0	A_z

$z + i$

Exam/Feb

Paper 4 of 7 — Phy 313 Past questions & Solution

Level 300 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria.
College of Science
Department of Physics
First Semester Examination

Academic Session: 2019/2020
Course Title: Analytical Mechanics I
Course Code: PHY313 (3 units)
Time Allowed: 3hrs.

27/02/2021

Instructions: Answer Question 1 and Any other Three Questions. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

Question 1

- A. Prove that the total angular momentum of a system of particles, referred to a given origin, can be expressed as the sum of two parts: one part being the angular momentum of all the particles relative to the centre-of-mass, and the other being the angular momentum of the total mass assumed to be at the centre-of-mass, relative to the origin.
- B. A system of particles consist of four particles of masses one quarter ($\frac{1}{4}$) unit each, with position vectors $\vec{r}_1 = 4t\hat{a}_x - 8t\hat{a}_y - 4t^2\hat{a}_z$, $\vec{r}_2 = 12t\hat{a}_x + 4t\hat{a}_y$, $\vec{r}_3 = \vec{r}_1$, $\vec{r}_4 = \vec{r}_2$, where t is the time. Calculate the total angular momentum of the system.

(22 marks)

Question 2

- A. (i) State the Divergence theorem in Cartesian coordinates, and define all terms.
(ii) Given a force $\vec{F} = x^2\hat{a}_x + xy\hat{a}_y + yz\hat{a}_z$, in Cartesian coordinate. Verify the divergence theorem over a cube half ($1/2$) unit on each side. The cube is located at the first octant of the Cartesian system with one corner at the origin.
- B. State the curl of a vector field in Cartesian coordinates, and explain the physical meaning of the expression. (ii) Show that $\text{curl } \vec{B} = 0$, if $\vec{B} = (k/r)\hat{a}_r$, in spherical coordinates, where k is a constant.

(16 marks)

Question 3

- A. Consider a particle moving in a spherical coordinate system. Write down the coordinate transformation equations and obtain the position vector, $\vec{r}$ in spherical coordinate.
- B. Derive an expression for the kinetic energy and potential energy possess by the particle.
- C. Hence, obtain the Lagrangian formulations and derive the Larange's equations for the three generalized coordinates.

(16 marks)

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Question 4

A. Let (r, θ) represents the plane polar coordinates describing the position vector of a particle P. If $\hat{r}$ is a unit vector in the direction of the position vector $\vec{r}$, and $\hat{\theta}$ is a unit vector perpendicular to $\hat{r}$ and in the direction of increasing θ .

- Derive an expression for the velocity of the particle.
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B. Assuming a particle is under the influence of a central force field $f(r) = \left(\frac{\alpha}{r^2} + \frac{\beta}{r^3} \right)$, where $\alpha, \beta > 0$. Find the orbit or path of the particle.

(16 marks)

Question 5

- Define central force motion and state three characteristics.
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Define the terms v, μ, a, r as it appears in C(i) above.

(16 marks)

Question 6

- Write down the point and vector transformations from (i) Cartesian to Spherical coordinates (ii) Spherical to Cartesian coordinates.
- (i) Given that vector $\vec{A} = A_r \hat{a}_r + A_\theta \hat{a}_\theta + A_z \hat{a}_z$, derive the transformation equation from Cylindrical to Cartesian coordinates in matrix form.

(16 marks)

$$\frac{1}{r} \propto \frac{1}{2} = \frac{1}{2}$$

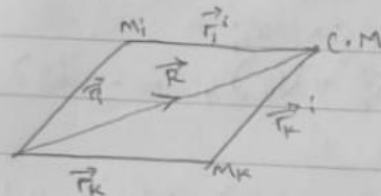
$$\frac{1}{2}$$

PHY 313.

2019/2020 Past Questions.

- 1(a) Prove that the total angular momentum of a system of particles, referred to a given origin can be expressed as the sum of 2 parts: one part being the angular momentum of all particles relative to the C.M and the other being the angular momentum of the total mass assumed to be at the C.M, relative to the origin.

solu.



The position of particle k with respect to the origin of mass is specified by the vector;

$$\vec{r}_k = \vec{R} + \vec{r}_k^i \quad \text{--- (1)} \quad k=1, 2, 3, \dots, N$$

from the definition of C.M;

$$M\vec{R} = \sum_{k=1}^N m_k \vec{r}_k \quad \text{--- (2)}$$

$$\left[\begin{array}{l} \vec{R} = \text{Position of C.M} \\ \vec{r}_k^i = \text{Internal coord} \\ \text{vector particle } k \end{array} \right]$$

The Internal coord $\vec{r}_k^i$ satisfy the equation;

$$\sum_{k=1}^N m_k \vec{r}_k^i = 0 \quad \text{--- (3)}$$

$$\sum_{k=1}^N m_k \vec{v}_k^i = 0 \quad \text{--- (*)}$$

we define the C.M velocity and the Internal

coordinates

velocities as;

$$\vec{R} = \frac{d\vec{R}}{dt} = \vec{V} \quad \text{--- (4)}$$

$$\vec{r}_k = \vec{R} + \vec{r}_k^i \quad \text{--- (5)}$$

$$\vec{v}_k = \vec{V} + \vec{v}_k^i \quad \text{--- (6)}$$

$$\vec{v}_k^i = \vec{v}_k - \vec{V} \quad \text{--- (7)}$$

Let us define the total angular momentum of the system of particles about the origin as;

$$\vec{L} = \sum_{k=1}^N m_k (\vec{r}_k \times \vec{v}_k) \quad \text{--- (8)}$$

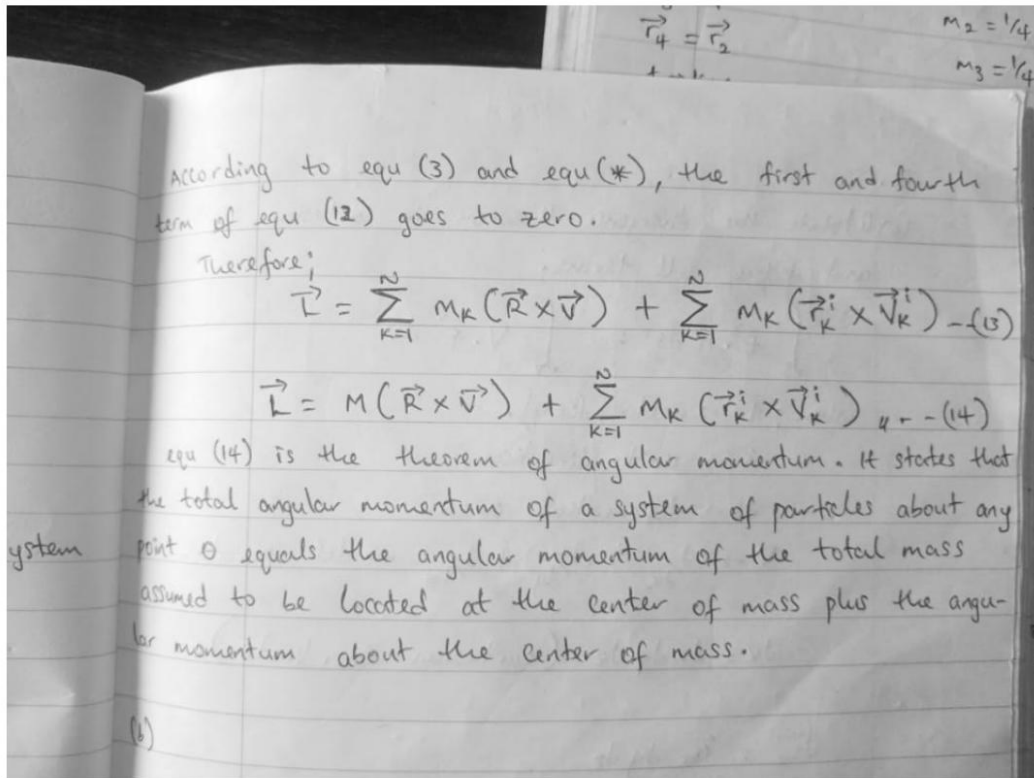
Substituting equ (1) and equ (6) into equ (8);

$$\vec{L} = \sum_{k=1}^N m_k [(\vec{R} + \vec{r}_k^i) \times (\vec{V} + \vec{v}_k^i)] \quad \text{--- (9)}$$

$$\vec{L} = \sum_{k=1}^N m_k [\vec{R} \times (\vec{v}_k^i + \vec{V}) + \vec{r}_k^i \times (\vec{v}_k^i + \vec{V})] \quad \text{--- (10)}$$

$$\begin{aligned} \vec{L} = & \sum_{k=1}^N m_k (\vec{R} \times \vec{v}_k^i) + \sum_{k=1}^N m_k (\vec{R} \times \vec{V}) + \sum_{k=1}^N m_k (\vec{r}_k^i \times \vec{v}_k^i) \\ & + \sum_{k=1}^N m_k (\vec{r}_k^i \times \vec{V}) \quad \text{--- (11)} \end{aligned}$$

$$\begin{aligned} \vec{L} = & \sum_{k=1}^N \vec{R} \times (m_k \vec{v}_k^i) + \sum_{k=1}^N m_k (\vec{R} \times \vec{V}) + \sum_{k=1}^N m_k (\vec{r}_k^i \times \vec{v}_k^i) \\ & + \sum_{k=1}^N \vec{V} \times (m_k \vec{r}_k^i) \quad \text{--- (12)} \end{aligned}$$



1(b)

$$\vec{r}_1 = 4t \hat{a}_x - 8 \hat{a}_y - 4t^2 \hat{a}_z$$

$$\vec{r}_2 = 12t \hat{a}_x + 4t \hat{a}_y$$

$$m_1 = 1/4$$

$$m_2 = 1/4$$

$$m_3 = 1/4$$

$$m_4 = 1/4$$

$t \rightarrow \text{time}$

$$\vec{r}_3 = 4t \hat{a}_x - 8 \hat{a}_y - 4t^2 \hat{a}_z$$

$$\vec{r}_4 = 12t \hat{a}_x + 4t \hat{a}_y$$

but ; $R = \frac{\sum m_i \vec{r}_i}{\sum m_i} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + m_4 \vec{r}_4}{m_1 + m_2 + m_3 + m_4}$

$$= \frac{\sum m_i}{\sum m_i} = 1/4 + 1/4 + 1/4 + 1/4 = 1$$

$R = 1/4$

$$R = \frac{1}{4} \left[4t \hat{a}_x - 8 \hat{a}_y - 4t^2 \hat{a}_z \right] = \left[(t) \hat{a}_x - (2) \hat{a}_y - (t^2) \hat{a}_z \right]$$

$$+ \frac{1}{4} \left[12t \hat{a}_x + 4t \hat{a}_y \right] = \left[(3t) \hat{a}_x + (t) \hat{a}_y \right]$$

$$+ \frac{1}{4} \left[4t \hat{a}_x - 8 \hat{a}_y - 4t^2 \hat{a}_z \right] = \left[(t) \hat{a}_x - (2) \hat{a}_y - (t^2) \hat{a}_z \right]$$

$$+ \frac{1}{4} \left[12t \hat{a}_x + 4t \hat{a}_y \right] = \left[(3t) \hat{a}_x + (t) \hat{a}_y \right]$$

$$\vec{R} = \left[t \hat{a}_x - 2t^2 \hat{a}_y - t^3 \hat{a}_z + 3t \hat{a}_x + t \hat{a}_y + t \hat{a}_x - 2t^2 \hat{a}_y - t^3 \hat{a}_z + 3t \hat{a}_x + t \hat{a}_y \right]$$

$$\vec{R} = (8t) \hat{a}_x + (2t - 4) \hat{a}_y - (2t^2) \hat{a}_z$$

$$\vec{v} = \frac{d\vec{R}}{dt} = 8 \hat{a}_x + 2 \hat{a}_y - 4t \hat{a}_z$$

$$M = \sum_i m_i = m_1 + m_2 + m_3 + m_4 = 1$$

$$\text{Since } \vec{r}_k = \vec{R} + \vec{r}_k^i \Rightarrow \vec{r}_k^i = \vec{r}_k - \vec{R}$$

$$\vec{r}_1^i = \vec{r}_1 - \vec{R}$$

$$= 4t \hat{a}_x - 8 \hat{a}_y - 4t^2 \hat{a}_z - [8t \hat{a}_x + (2t - 4) \hat{a}_y - 2t^2 \hat{a}_z]$$

$$= 4t \hat{a}_x - 8t \hat{a}_x - (-8 - (2t - 4)) \hat{a}_y - 4t^2 + 2t^2 \hat{a}_z$$

$$-4t \hat{a}_x - 8 + 4 - 2t \hat{a}_y - 2t^2 \hat{a}_z$$

$$-4t \hat{a}_x - 4 - 2t \hat{a}_y - 2t^2 \hat{a}_z$$

$$\vec{r}_1^i = -4t \hat{a}_x - (4 + 2t) \hat{a}_y - 2t^2 \hat{a}_z$$

$$\vec{r}_2^i = \vec{r}_2 - \vec{R}$$

$$= 12t \hat{a}_x + 4t \hat{a}_y - [8t \hat{a}_x + (2t - 4) \hat{a}_y - 2t^2 \hat{a}_z]$$

$$= 12t \hat{a}_x - 8t \hat{a}_x + 4t - (2t - 4) \hat{a}_y + 2t^2 \hat{a}_z$$

$$4t \hat{a}_x + 4t - 2t + 4 \hat{a}_y + 2t^2 \hat{a}_z$$

$$\vec{r}_2^i = 4t \hat{a}_x + (2t + 4) \hat{a}_y + 2t^2 \hat{a}_z$$

$$\begin{aligned}\vec{r}_3^i &= \vec{r}_3 - \vec{R} \\ \text{for } \vec{r}_3 &= \vec{r}_1 \\ \vec{r}_3^i &= -4t \hat{a}_x - (4+2t) \hat{a}_y - 2t^2 \hat{a}_z \\ \text{for } \vec{r}_4 &= \vec{r}_2 \\ \vec{r}_4^i &= \vec{r}_4 - \vec{R} \\ \vec{r}_4^i &= 4t \hat{a}_x \\ \vec{r}_4^i &= 4t \hat{a}_x + (2t+4) \hat{a}_y + 2t^2 \hat{a}_z\end{aligned}$$

The velocities are;

$$\begin{aligned}\vec{v}_1^i &= \frac{d\vec{r}_1^i}{dt} = -4\hat{a}_x - 2\hat{a}_y - 4t\hat{a}_z \\ &= -(4\hat{a}_x + 2\hat{a}_y + 4t\hat{a}_z) \\ \vec{v}_2^i &= \frac{d\vec{r}_2^i}{dt} = 4\hat{a}_x + 2\hat{a}_y + 4t\hat{a}_z \\ \vec{v}_3^i &= \vec{v}_1^i = -(4\hat{a}_x + 2\hat{a}_y + 4t\hat{a}_z) \\ \vec{v}_4^i &= \vec{v}_2^i = 4\hat{a}_x + 2\hat{a}_y + 4t\hat{a}_z\end{aligned}$$

$$\vec{L} = M(\vec{R} \times \vec{v}) + \sum_{k=1}^N m_k (\vec{r}_k^i \times \vec{v}_k^i)$$

~~$M(\vec{R} \times \vec{v})$~~ $\hat{i} \quad \hat{j} \quad \hat{k}$

$$M(\vec{r} \times \vec{v}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 8t & (2t-4) & -2t^2 \\ 8 & 2 & -4t \end{vmatrix}$$

$$\hat{i} [(2t-4)(-4t) - (-2t^2)(2)] - \hat{j} [(8t)(-4t) - (-2t^2)(8)]$$

$$+ \hat{k} [(8t)(2) - (2t-4)(8)]$$

$$\hat{i} [-8t^2 + 16t - (-4t^2)] - \hat{j} [-32t^2 - (-16t^2)]$$

$$+ \hat{k} [16t - (16t - 32)]$$

$$\hat{i} [16t - 4t^2] - \hat{j} [-16t^2] + \hat{k} [32]$$

$$M(\vec{r} \times \vec{v}) = \hat{i} [16t - 4t^2] + \hat{j} [16t^2] + \hat{k} [32] \text{ u.s.}$$

similarly, we evaluate;

$$m_1(\vec{r}_1 \times \vec{v}_1), m_2(\vec{r}_2 \times \vec{v}_2), m_3(\vec{r}_3 \times \vec{v}_3)$$

$$m_4(\vec{r}_4 \times \vec{v}_4).$$

$$m_1(\vec{r}_1 \times \vec{v}_1) = \frac{1}{4} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -4t & -(4+2t) & -2t^2 \\ -4 & -2 & -4t \end{vmatrix}$$

$$\frac{1}{4} \left[\hat{i} [-(4+2t)(-4t) - (-2t^2)(-2)] - \hat{j} [(-4t)(-4t) - (-2t^2)(-4)] \right]$$

$$\begin{aligned}
 & + \hat{k} [(-4t)(-2) - (- (4+2t))(-4)] \\
 & = \frac{1}{4} \left[\hat{i} (16t + 8t^2) - (4t^2) - \hat{j} (16t^2 - 8t^2) + \hat{k} (8t - (16+8t)) \right] \\
 & \frac{1}{4} \left[\hat{i} (16t - 4t^2) - \hat{j} (8t^2) + \hat{k} (-16) \right] \\
 \\
 m_1 (\vec{r}_1^i \times \vec{v}_1^i) &= (4t - t^2) \hat{i} - (2t^2) \hat{j} - (4t) \hat{k} \\
 m_2 (\vec{r}_2^i \times \vec{v}_2^i) &= \frac{1}{4} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4t & (2t+4) & 2t^2 \\ 4 & 2 & 4t \end{vmatrix} \\
 & \frac{1}{4} \left[\hat{i} (2t+4)(4t) - (2t^2)(2) - \hat{j} (4t)(4t) - (2t^2)(4) \right. \\
 & \quad \left. + \hat{k} (4t)(2) - (2t+4)(4) \right] \\
 & \frac{1}{4} \left[\hat{i} (8t^2 + 16t) - (4t^2) - \hat{j} (16t^2 - 8t^2) + \hat{k} (8t - (8t+16)) \right] \\
 & \frac{1}{4} \left[\hat{i} (4t^2 + 16t) - \hat{j} (8t^2) + \hat{k} (-16) \right]
 \end{aligned}$$

$$\begin{aligned}
 m_1 (\vec{r}_1^i \times \vec{v}_1^i) &= [4t - t^2] \hat{i} - [2t^2] \hat{j} - [4] \hat{k} \\
 m_2 (\vec{r}_2^i \times \vec{v}_2^i) &= [(t^2 + 4t) \hat{i} - (2t^2) \hat{j} - (4) \hat{k}] \\
 m_3 (\vec{r}_3^i \times \vec{v}_3^i) &= [(t^2 + 4t) \hat{i} - (2t^2) \hat{j} - (4) \hat{k}] \\
 m_4 (\vec{r}_4^i \times \vec{v}_4^i) &= [(t^2 + 4t) \hat{i} - (2t^2) \hat{j} - (4) \hat{k}] \\
 \sum_{k=1}^N m_k (\vec{r}_k^i \times \vec{v}_k^i) &= \left[[4t - t^2] + [t^2 + 4t] + [t^2 + 4t] + [t^2 + 4t] \right] \hat{i} \\
 &+ \left[(-2t^2) + (-2t^2) + (-2t^2) + (-2t^2) \right] \hat{j} \\
 &+ (-4) + (-4) + (-4) + (-4) \hat{k} \\
 \sum_{k=1}^N m_k (\vec{r}_k^i \times \vec{v}_k^i) &= \left[(4t - t^2 + t^2 + 4t + t^2 + 4t + t^2 + 4t) \hat{i} \right. \\
 &\left. (-2t^2 - 2t^2 - 2t^2 - 2t^2) \hat{j} \right. \\
 &\left. (-4 - 4 - 4 - 4) \hat{k} \right] \\
 \sum_{k=1}^N m_k (\vec{r}_k^i \times \vec{v}_k^i) &= 16t \hat{i} - 8t^2 \hat{j} - 16 \hat{k} \\
 \vec{L} &= M (\vec{R} \times \vec{V}) + \sum_{k=1}^N m_k (\vec{r}_k^i \times \vec{v}_k^i) \\
 &= [(16t - 4t^2) \hat{i} + 16t^2 \hat{j} + 32 \hat{k}] + [16t \hat{i} - 8t^2 \hat{j} - 16 \hat{k}]
 \end{aligned}$$

center of mass plus the angular

$$\vec{L} = [(82t - 4t^2)\hat{i} + (8t^2)\hat{j} + (16)\hat{k}] \text{ u.}$$

2.

(a) (i) state the divergence theorem in Cartesian coordinates and define all terms.

$$\int_S \vec{F} \cdot \hat{n} \, ds = \int_V \nabla \cdot \vec{F} \, dv$$

where; $\vec{F} \rightarrow$ vector field.

$\hat{n} \rightarrow$ unit direction.

$ds \rightarrow$ differential area.

$\nabla \rightarrow \frac{d}{dx} + \frac{d}{dy} + \frac{d}{dz} \rightarrow$ del operator.

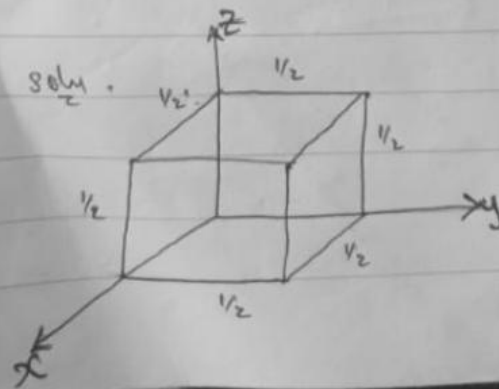
$dv = dx \, dy \, dz$ (Cartesian coordinate).

$$dS_x = \hat{a}_x \, dy \, dz$$

$$dS_y = \hat{a}_y \, dx \, dz$$

$$dS_z = \hat{a}_z \, dx \, dy.$$

(ii) For $\vec{F} = x^2 \hat{a}_x + xy \hat{a}_y + yz \hat{a}_z$, verify the divergence theorem over a cube half ($\frac{1}{2}$) unit on each side.



LHS.

front face $\Rightarrow x = \frac{1}{2}; d\vec{s}_x = \hat{a}_x dy dz.$

$$\int_S \vec{F} \cdot d\vec{s} = \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} (\hat{a}_x x^2 + \hat{a}_y xy + \hat{a}_z yz) \cdot (\hat{a}_x dy dz)$$

$$\int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} x^2 dy dz = \int_0^{\frac{1}{2}} [x^2 y]_0^{\frac{1}{2}} dz$$

$$= \left[x^2 y z \right]_0^{\frac{1}{2}} \bigg|_0^{\frac{1}{2}}$$

$$x = \frac{1}{2}$$

$$= \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = \frac{1}{16}.$$

Back face $\Rightarrow x = 0; d\vec{s}_x = -\hat{a}_x dy dz$

$$\int \vec{F} \cdot d\vec{s} = \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} (-x^2) dy dz = 0 \text{ since } x = 0.$$

Right face $\Rightarrow y = \frac{1}{2}; d\vec{s}_y = \hat{a}_y dx dz.$

$$\int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} (\hat{a}_x x^2 + \hat{a}_y xy + \hat{a}_z yz) \cdot (\hat{a}_y dx dz)$$

$$\int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} xy dx dz = \int_0^{\frac{1}{2}} \left[\frac{x^2 y}{2} \right]_0^{\frac{1}{2}} dz$$

$$= \left[\frac{x^2 y z}{2} \right]_0^{\frac{1}{2}} \bigg|_0^{\frac{1}{2}}$$

$$= \frac{(\frac{1}{2})^2 (\frac{1}{2}) (\frac{1}{2})}{2} = \frac{1}{32}$$

Left face; $y = 0; d\vec{s}_y = -\hat{a}_y dx dz$

$$\int \vec{F} \cdot d\vec{s} = \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} (-xy) dx dz = 0 \text{ since } y = 0.$$

Top face; $z = \frac{1}{2}$; $d\vec{S} = \hat{a}_z dx dy$

$$\int_S \vec{F} \cdot d\vec{S} = \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} (x^2 \hat{a}_x + xy \hat{a}_y + yz \hat{a}_z) \cdot (\hat{a}_z dx dy)$$

$$= \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} yz dx dy$$

$$= \int_0^{\frac{1}{2}} xy z \Big|_0^{\frac{1}{2}} dy$$

$$= \left[\frac{xy^2 z}{2} \Big|_0^{\frac{1}{2}} \right]_0^{\frac{1}{2}}$$

$$= \frac{(\frac{1}{2})(\frac{1}{2})^2(\frac{1}{2})}{2} = \frac{1}{32}$$

$$\int_S \vec{F} \cdot d\vec{S} = \frac{1}{16} + \frac{1}{32} + \frac{1}{32} = \frac{1}{8} \text{ u.}$$

RHS; $dv = dx dy dz$,

$$\oint \nabla \cdot \vec{F} dv = \int_V (\hat{a}_x \frac{d}{dx} + \hat{a}_y \frac{d}{dy} + \hat{a}_z \frac{d}{dz}) \cdot (\hat{a}_x x^2 + \hat{a}_y xy + \hat{a}_z yz) dx dy dz$$

$$= \int_V \frac{d}{dx}(x^2) + \frac{d}{dy}(xy) + \frac{d}{dz}(yz) dx dy dz$$

$$= \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} (2x + x + y) dx dy dz$$

$$\int_V \left(\frac{2x^2}{2} + \frac{x^2}{2} + xy \right) dy dz$$

$$\int_V \left(\frac{2x^2 y}{2} + \frac{x^2 y}{2} + \frac{xy^2}{2} \right) dz$$

$$\int_0^1 \left[\frac{2x^2 y z}{2} + \frac{x^2 y z}{2} + \frac{xy^2 z}{2} \right]_{z=0}^{z=1/2} dx$$

$$\frac{x \left(\frac{1}{2} \right)^2 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)}{2} + \frac{\left(\frac{1}{2} \right)^2 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)}{2} + \frac{\left(\frac{1}{2} \right) \left(\frac{1}{2} \right)^2 \left(\frac{1}{2} \right)}{2}$$

$$= \frac{1}{16} + \frac{1}{32} + \frac{1}{32} = \frac{1}{8}$$

$$\int_S \vec{F} \cdot d\vec{s} = \int_V \nabla \cdot \vec{F} dv = \frac{1}{8}$$

This verifies the divergence theorem over a cube in Cartesian coordinate system.

$$(b). \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

$$\hat{i} \left[\frac{\partial(A_z)}{\partial y} - \frac{\partial(A_y)}{\partial z} \right] - \hat{j} \left[\frac{\partial(A_z)}{\partial x} - \frac{\partial(A_x)}{\partial z} \right] + \hat{k} \left[\frac{\partial(A_y)}{\partial x} - \frac{\partial(A_x)}{\partial y} \right]$$

The physical meaning depends on what kind of field vector $\vec{F}$ represents.

(i) show that $\text{curl } \vec{B} = 0$. If $\vec{B} = (K/r) \hat{a}_r$ in spherical coordinates where K is a constant.

Soln.

$$\nabla \times \vec{B} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{a}_r & r \hat{a}_\theta & r \sin \theta \hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ K & 0 & 0 \end{vmatrix}$$

$$\frac{1}{r^2 \sin \theta} \left[r \hat{a}_\phi \left[\frac{\partial}{\partial \theta} (0) - \frac{\partial}{\partial \phi} (0) \right] - r \hat{a}_\theta \left[\frac{\partial}{\partial r} (0) - \frac{\partial}{\partial \phi} (K) \right] + r \sin \theta \hat{a}_\phi \left[\frac{\partial}{\partial r} (0) - \frac{\partial}{\partial \theta} (K) \right] \right]$$

differentiating a constant gives 0.

$$\frac{1}{r^2 \sin \theta} (0) = 0$$

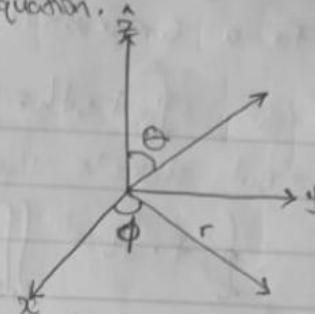
$$\therefore \nabla \times \vec{B} = 0$$

3-(a) Coordinate transformation equation.

$$z = r \cos \theta$$

$$y = r \sin \theta \sin \phi$$

$$x = r \sin \theta \cos \phi$$



Position vector in spherical coordinate.

$$\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$$

$$\vec{r} = r \sin \theta \cos \phi \hat{i} + r \sin \theta \sin \phi \hat{j} + r \cos \theta \hat{k}$$

$$\vec{r} = r (\sin \theta \cos \phi \hat{i} + \sin \theta \sin \phi \hat{j} + \cos \theta \hat{k})$$

$$\hat{r} = \frac{d\vec{r}}{dr} = \sin \theta \cos \phi \hat{i} + \sin \theta \sin \phi \hat{j} + \cos \theta \hat{k}$$

$$\vec{r} = r \hat{r}(\theta, \phi)$$

3b. $dw = F dl$

$$\int_a^b dw = \int_a^b F dl$$

$$= \int_a^b m a dl = m \int_a^b \frac{dv}{dt} dl$$

$$= m \int_a^b \frac{dl}{dt} dv = m \int_b^a v dv$$

$$= m \left[\frac{v^2}{2} \right]_a^b = \frac{1}{2} m [v_b^2 - v_a^2]$$

To get the Potential energy;
For a Conservative force;

$$\oint_C \mathbf{F} \cdot d\mathbf{l} = 0$$

But Stoke's theorem;

$$\oint_C \mathbf{F} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{n} \, ds = 0$$

$$\nabla \times \mathbf{F} = 0$$

This holds when; $\mathbf{F} = -\nabla\phi$

$$\mathbf{F} = -\frac{d\phi}{dr} \quad r \rightarrow (x, y, z).$$

$$\mathbf{F} \cdot d\mathbf{r} = -d\phi$$

$$\int \mathbf{F} \cdot d\mathbf{r} = -\int d\phi$$

$$\int (mg) \cdot d\mathbf{r} = -\int d\phi$$

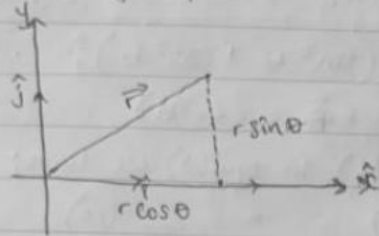
$$mg \int_a^b dr = -\int_a^b d\phi$$

$$[mgr]_a^b = -[\phi]_a^b$$

or;

$$\phi = -mgr.$$

4. Let (r, θ) represent the plane polar coordinates describing the position vector of a particle P.



$$\vec{r} = r \cos \theta \hat{i} + r \sin \theta \hat{j} \quad \text{---(1)}$$

$$\vec{r} = r (\cos \theta \hat{i} + \sin \theta \hat{j}) \quad \text{---(2)}$$

$$\hat{r} = \frac{d\vec{r}}{dr} = \cos \theta \hat{i} + \sin \theta \hat{j} \quad \text{---(3)}$$

$$\hat{r}(\theta) = \cos \theta \hat{i} + \sin \theta \hat{j} \quad \text{---(4)}$$

$$\hat{\theta} = \frac{d\hat{r}}{d\theta} = -\sin \theta \hat{i} + \cos \theta \hat{j} \quad \text{---(5)}$$

$$\hat{\theta}(\theta) = -\sin \theta \hat{i} + \cos \theta \hat{j} \quad \text{---(6)}$$

$$\frac{d\hat{\theta}}{d\theta} = -\cos \theta \hat{i} - \sin \theta \hat{j} \quad \text{---(7)}$$

$$\frac{d\hat{\theta}}{d\theta} = -(\cos \theta \hat{i} + \sin \theta \hat{j}) = -\hat{r} \quad \text{---(8)}$$

Let us consider equ (3) and equ (5):

$$\hat{r} = \cos \theta \hat{i} + \sin \theta \hat{j} \quad \times \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \times \sin \theta$$

$$\hat{\theta} = -\sin \theta \hat{i} + \cos \theta \hat{j} \quad \times \begin{bmatrix} \sin \theta \\ \cos \theta \end{bmatrix} \times \cos \theta$$

$$\cos \theta \hat{r} = \cos^2 \theta \hat{i} + \sin \theta \cos \theta \hat{j} \quad \text{---(9)}$$

$$-\sin \theta \hat{\theta} = -\sin^2 \theta \hat{i} + \sin \theta \cos \theta \hat{j} \quad \text{---(10)}$$

$$\cos \theta \hat{r} - \sin \theta \hat{\theta} = \cos^2 \theta \hat{i} + \sin^2 \theta \hat{j}$$

$$\hat{j} = \cos \theta \hat{r} - \sin \theta \hat{\theta} \quad \text{---(11)}$$

$$\sin \theta \hat{r} = \sin \theta \cos \theta \hat{i} + \sin^2 \theta \hat{j} \quad \text{--- (12)}$$

$$\cos \theta \hat{\theta} = -\sin \theta \cos \theta \hat{i} + \cos^2 \theta \hat{j} \quad \text{--- (13)}$$

$$\sin \theta \hat{r} + \cos \theta \hat{\theta} = (\sin^2 \theta + \cos^2 \theta) \hat{j}$$

$$\hat{j} = \sin \theta \hat{r} + \cos \theta \hat{\theta} \quad \text{--- (14)}$$

we can replace equ (14) in equ (2) we have;

$$\vec{r} = r \hat{r}(\theta) \quad \text{--- (15)}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \dot{r} = r \frac{d\hat{r}}{dt} + \hat{r} \frac{dr}{dt} \quad \text{--- (16)}$$

$$= r \left[\frac{d\hat{r}}{d\theta} \times \frac{d\theta}{dt} \right] + \hat{r} \dot{r}$$

$$= r (\hat{\theta}) (\dot{\theta}) + \hat{r} \dot{r} \quad \text{--- (17)}$$

$$r \dot{\theta} \hat{\theta} + \hat{r} \dot{r}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \ddot{r} = r \dot{\theta} \frac{d\hat{\theta}}{dt} + r \hat{\theta} \frac{d\dot{\theta}}{dt} + \hat{\theta} \dot{\theta} \frac{dr}{dt} + \hat{r} \frac{d\dot{r}}{dt} + \hat{r} \frac{d\dot{r}}{dt} \quad \text{--- (18)}$$

$$= r \dot{\theta} \left[\frac{d\hat{\theta}}{d\theta} \times \frac{d\theta}{dt} \right] + r \hat{\theta} \ddot{\theta} + \hat{\theta} \dot{\theta} \dot{r} + \hat{r} \left[\frac{d\hat{r}}{d\theta} \times \frac{d\theta}{dt} + \hat{r} \ddot{r} \right]$$

$$= r \dot{\theta} (-\hat{r}) \dot{\theta} + r \ddot{\theta} \hat{\theta} + \hat{\theta} \dot{\theta} \dot{r} + \hat{r} (\hat{\theta}) \dot{\theta} + \ddot{r} \hat{r} \quad \text{--- (19)}$$

$$= -r \dot{\theta}^2 \hat{r} + r \ddot{\theta} \hat{\theta} + 2 \hat{r} \dot{\theta} \dot{r} + \ddot{r} \hat{r}$$

$$\vec{a} = (\ddot{r} - r \dot{\theta}^2) \hat{r} + (r \ddot{\theta} + 2 \dot{r} \dot{\theta}) \hat{\theta} \quad \text{--- (20)}$$

There is no force acting in the direction of the angular acceleration, since the force is a central force always in the direction of the radial acceleration.

(mass) (acceleration) = net force $\rightarrow (21)$

$$m \left[\left[\frac{d^2 r}{dt^2} - r \left(\frac{d\dot{\theta}}{dt} \right)^2 \right] \hat{r} + \frac{1}{r} \frac{d}{dt} \left(r^2 \frac{d\dot{\theta}}{dt} \right) \hat{\theta} \right] = f(r) \hat{r}$$

$$\frac{d}{dt} (r^2 \dot{\theta}) = \frac{d(r^2)}{dt} \dot{\theta} + r^2 \frac{d\dot{\theta}}{dt} = \frac{d}{dr} (r^2) \cdot \frac{dr}{dt} \dot{\theta} + r^2 \frac{d\dot{\theta}}{dt} \quad \rightarrow (22)$$

$$= 2r\dot{r}\dot{\theta} + r^2 \ddot{\theta} \quad \rightarrow (23)$$

required equations of motion are;

$$m(\ddot{r} - r\dot{\theta}^2) = f(r) \quad \rightarrow (24)$$

$$m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = 0 \quad \rightarrow (25)$$

from equ; (25);

$$m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = \frac{m}{r} (r^2 \ddot{\theta} + 2r\dot{r}\dot{\theta}) = 0 \quad \rightarrow (26)$$

but; $\frac{m}{r} \neq 0$

$$r^2 \ddot{\theta} + 2r\dot{r}\dot{\theta} = 0 \quad \rightarrow (27)$$

equ (27) can also be re-written as;

$$\frac{d}{dt} (r^2 \dot{\theta}) = 0 \quad \rightarrow (28)$$

Integrating through;

$$\int \frac{d}{dt} r^2 \dot{\theta} = \int 0$$

$$r^2 \dot{\theta} = h \quad ; \quad h \rightarrow \text{Constant.}$$

$$\dot{\theta} = \frac{h}{r^2}$$

$$\dot{\theta}^2 = \left(\frac{h}{r^2} \right)^2 = \frac{h^2}{r^4} \quad \rightarrow (29)$$

Put equ (29) into equ (24);

$$m(\ddot{r} - r(\dot{\theta}^2/r^4)) = f(r) \quad \dots (30)$$

$$m(\ddot{r} - \dot{\theta}^2/r^3) = f(r) \quad \dots (31)$$

$$\text{let } r = \frac{1}{u} \quad \dots *$$

$$\text{from } (*) \quad \left[\begin{array}{l} dr = 1 \\ u^2 r^2 = 1 \\ u^3 r^3 = 1 \end{array} \right] \quad \dots (32)$$

$$\frac{dr}{du} = \frac{d}{du} \left(\frac{1}{u} \right)$$

$$\frac{dr}{du} = -\frac{1}{u^2} \quad \dots (33)$$

for; $\dot{r}$

$$\dot{r} = \frac{dr}{dt} = \frac{dr}{d\theta} \cdot \frac{d\theta}{dt} = \dot{\theta} \frac{dr}{d\theta}$$

$$\text{recall that } \dot{\theta} = h/r^2$$

$$\dot{r} = \frac{dr}{dt} = \left(\frac{h}{r^2} \right) \frac{dr}{d\theta} \quad \dots (34)$$

$$\dot{r} = \frac{h}{r^2} \frac{dr}{d\theta} = \frac{h}{r^2} \frac{dr}{du} \cdot \frac{du}{d\theta} = \frac{h}{r^2} \left[-\frac{1}{u^2} \cdot \frac{du}{d\theta} \right]$$

$$= \frac{-h}{u^2 r^2} \cdot \frac{du}{d\theta}$$

recall that; $u^2 r^2 = 1$

$$\dot{r} = -h \frac{du}{d\theta} \quad \dots (35)$$

and;

$$\ddot{r} = \frac{dr}{dt} = m \frac{d}{dt} \left(\frac{dr}{dt} \right) = \frac{d}{dt} \left(-h \frac{du}{d\theta} \right) \quad \text{--- (36)}$$

$$= -h \frac{d}{d\theta} \left[\frac{du}{d\theta} \cdot \frac{d\theta}{dt} \right] = -h \dot{\theta} \frac{d^2 u}{d\theta^2}$$

$$= -h \left[\frac{h}{r^2} \right] \frac{d^2 u}{d\theta^2} = -\frac{h^2}{r^2} \frac{d^2 u}{d\theta^2} \quad \text{--- (37)}$$

$$\ddot{r} = -\frac{h^2}{r^2} \cdot \frac{u^2}{u^2} \frac{d^2 u}{d\theta^2} = -\frac{h^2 u^2}{r^2 u^2} \frac{d^2 u}{d\theta^2}$$

recall that; $r^2 u^2 = 1$

$$\ddot{r} = -h^2 u^2 \frac{d^2 u}{d\theta^2} \quad \text{--- (38)}$$

now we put equ (38) into equ (31).

$$m \left[-h^2 u^2 \frac{d^2 u}{d\theta^2} - \frac{h^2}{r^3} \right] = f(r).$$

$$m \left[-h^2 u^2 \frac{d^2 u}{d\theta^2} - \frac{h^2 \cdot u^3}{r^3 u^3} \right] = f(r) \quad \text{--- (39)}$$

for; $u^3 r^3 = 1$ and $r = 1/u$

$$m \left[-h^2 u^2 \frac{d^2 u}{d\theta^2} - h^2 u^3 \right] = f(1/u).$$

$$mh^2 u^2 \left[\frac{d^2 u}{d\theta^2} + u \right] = -f(1/u).$$

$$f(1/u) = -mh^2 u^2 \left[u + \frac{d^2 u}{d\theta^2} \right] \quad \text{--- (40)}$$

(4b). Assuming a Particle is under the Influence of a Central force field $f(r) = \left(\frac{\alpha}{r} + \frac{\beta}{r}\right)$.

solve.

$$f(r) = \left[\frac{\alpha}{r} + \frac{\beta}{r}\right]$$

recall; $\hat{r} f(r) = m(\ddot{r} - r\dot{\theta}^2)\hat{r} + m(2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}$ - (1)

by equating components;

$$m(\ddot{r} - r\dot{\theta}^2) = f(r) \quad - (2)$$

$$m(2\dot{r}\dot{\theta} + r\ddot{\theta}) = 0 \quad - (3)$$

let; $r = \frac{1}{u}$.

$$\frac{dr}{dt} = \dot{r} = \frac{dr}{du} \cdot \frac{du}{dt} = -\frac{1}{u^2} \frac{du}{dt} \quad - (4)$$

$$= -\frac{1}{u^2} \frac{du}{d\theta} \cdot \frac{d\theta}{dt} = -\frac{1}{u^2} \dot{\theta} \frac{du}{d\theta} \quad - (5)$$

since; $\dot{\theta} = \frac{h}{r^2} = \dot{\theta} \frac{h}{(1/u)^2} = hu^2 \quad - (6)$

$$\dot{r} = -\frac{1}{u^2} \cdot hu^2 \frac{du}{d\theta} = -h \frac{du}{d\theta} \quad - (7)$$

$$\ddot{r} = -h \frac{d}{dt} \left(\frac{du}{d\theta} \right) = -h \frac{d}{d\theta} \left(\frac{du}{dt} \right) = -h \frac{d}{d\theta} \left[\frac{du}{d\theta} \cdot \frac{d\theta}{dt} \right]$$

$$= -h \frac{d}{d\theta} \left[\frac{du}{d\theta} \dot{\theta} \right] = -h (hu^2) \frac{d^2u}{d\theta^2} \quad - (8)$$

$$\ddot{r} = -h^2 u^2 \frac{d^2u}{d\theta^2} \quad - (9)$$

Also; $f(r) = \left[\frac{\alpha}{r} + \frac{\beta}{r} \right]$

$$f\left(\frac{1}{u}\right) = \left[\frac{\alpha}{r} + \frac{\beta}{r} \right] \quad \text{--- (10)}$$

substituting equ (6) and equ (10) into equ (2);

$$m \left[-h^2 u^2 \frac{d^2 u}{d\theta^2} - \frac{1}{u} (h u^2)^2 \right] = \left[\frac{\alpha}{r} + \frac{\beta}{r} \right] \quad \text{--- (11)}$$

since $r = 1/u$.

$$m \left[-h^2 u^2 \frac{d^2 u}{d\theta^2} - \frac{1}{u} (h u^2)^2 \right] = [\alpha u + \beta u] \quad \text{--- (12)}$$

$$-m h^2 u^2 \left[\frac{d^2 u}{d\theta^2} + u \right] = u [\alpha + \beta] \quad \text{--- (13)}$$

$$\frac{d^2 u}{d\theta^2} + u = \frac{u [\alpha + \beta]}{-m h^2 u^2} \quad \text{--- (14)}$$

$$\left[\frac{d^2 u}{d\theta^2} + u = \frac{\alpha + \beta}{-m h^2 u} \right] \quad \text{--- (15)}$$

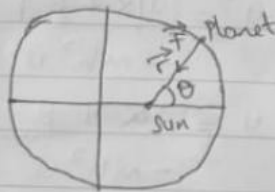
5(a) Central force motion is a force that is directed towards or away from a point called centre of force.

* characteristics

- i) The motion takes place in a plane.
- ii) The angular momentum about the origin is constant.
- iii) The particle describes equal area in equal times; $\vec{r} = \frac{d\mathbf{A}}{dt}$.

(b). Kepler's laws:

- i) Every planet moves in an orbit which is an ellipse with the Sun at one ~~foci~~ foci.
- ii) A radius vector joining any planet to the Sun sweeps out equal areas in equal lengths of time.
- iii) The squares of the Periods of revolution of the planets are proportional to the cubes of the semi-major axis of their orbits.



$$F = \frac{mv^2}{r} = \frac{Gmm}{r^2}$$

$$\left(\frac{mv^2}{r} = \frac{Gmm}{r^2} \right) \times \frac{1}{m}$$

$$\left(\frac{v^2}{r} = \frac{Gm}{r^2} \right) \times r$$

$$v^2 = \frac{Gm}{r}$$

$$\left(\frac{\Delta x}{\Delta t} \right)^2 = \frac{Gm}{r}$$

$\Delta x \rightarrow$ circumference of the circle.

$\Delta t \rightarrow$ Time taken to complete one oscillation. (T)

$$\left(\frac{2\pi r}{T} \right)^2 = \frac{Gm}{r}$$

$$\frac{4\pi^2 r^2}{T^2} = \frac{Gm}{r}$$

$$\frac{T^2}{4\pi^2 r^2} = \frac{r}{Gm}$$

$$T^2 = \frac{4\pi^2 r^3}{Gm}$$

$$T^2 = \left(\frac{4\pi^2}{Gm} \right) r^3$$

$$T^2 \propto r^3$$

This is a Prove of ~~Kepler~~ ^{4.} Kepler's third law.

6.(a) Cartesian $\rightarrow$ Spherical.

Point transformation.

$$r = (x^2 + y^2 + z^2)^{1/2}$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right)$$

$$\phi = \tan^{-1} (y/x)$$

vector transformation.

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Spherical $\rightarrow$ Cartesian.

Point transformation.

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

vector transformation.

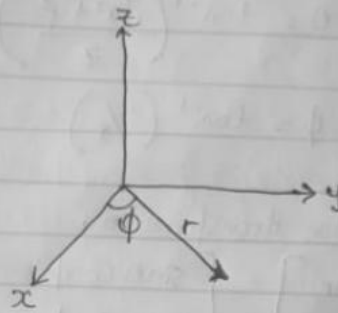
$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

6 (b) Given that a vector field $\vec{A} = \hat{a}_r A_r + \hat{a}_\phi A_\phi + \hat{a}_z A_z$, derive the transformation equation from cylindrical to Cartesian coordinates in matrix form.

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$z = z.$$



$$\vec{r} = x \hat{i} + y \hat{j}$$

$$\vec{r} = r \cos \phi \hat{i} + r \sin \phi \hat{j}$$

$$\hat{r} = \frac{d\vec{r}}{dr} = \cos \phi \hat{i} + \sin \phi \hat{j} \quad \text{--- (1)}$$

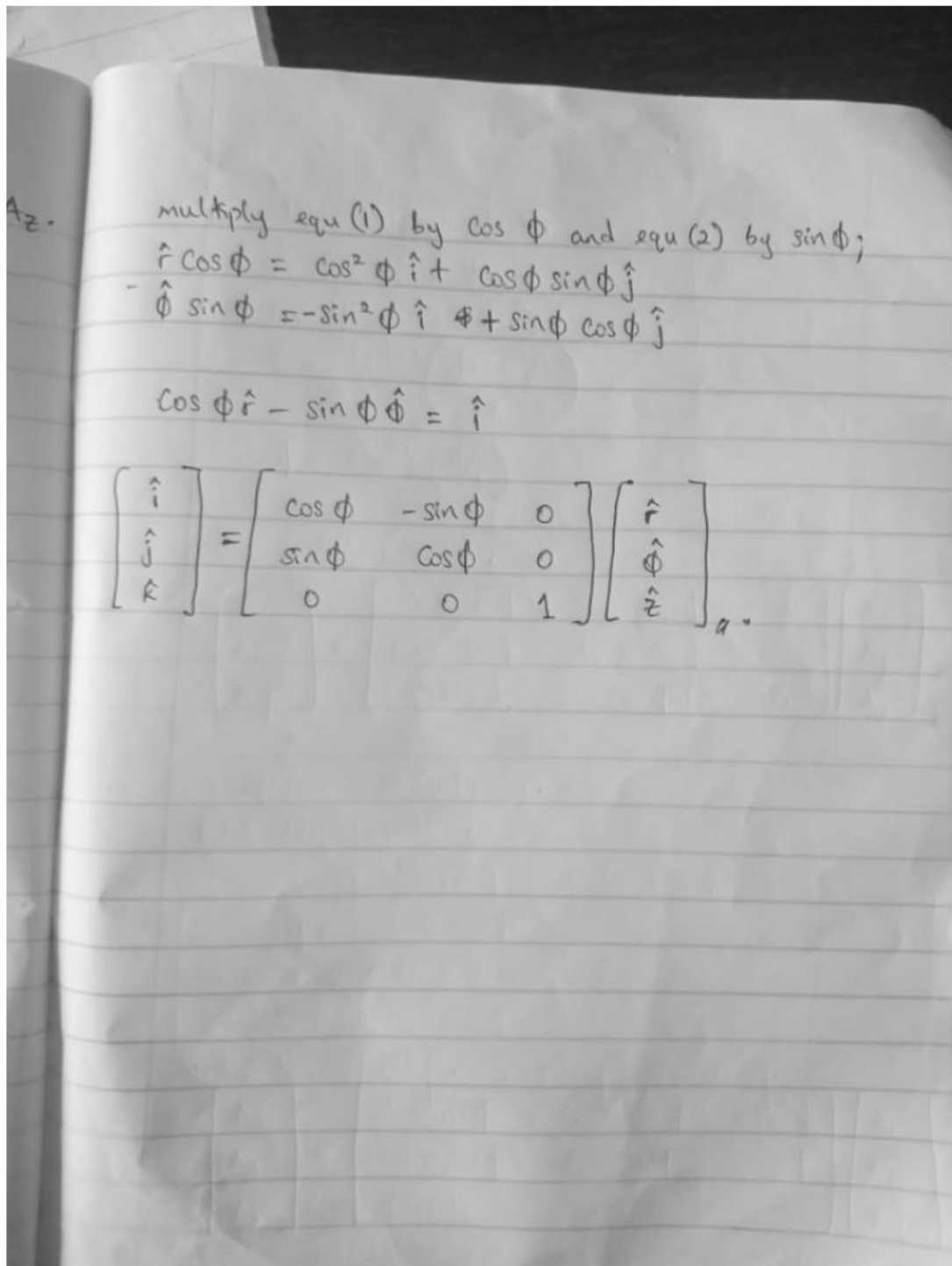
$$\hat{\phi} = \frac{d\hat{r}}{d\phi} = -\sin \phi \hat{i} + \cos \phi \hat{j} \quad \text{--- (2)}$$

multiply equ (1) by $\sin \phi$ and equ (2) by $\cos \phi$;

$$\hat{r} \sin \phi = \cos \phi \sin \phi \hat{i} + \sin^2 \phi \hat{j} \quad \text{--- (3)}$$

$$+ \hat{\phi} \cos \phi = -\sin \phi \cos \phi \hat{i} + \cos^2 \phi \hat{j} \quad \text{--- (4)}$$

$$\sin \phi \hat{r} + \cos \phi \hat{\phi} = \hat{j}$$



Paper 5 of 7 — Phy 313 Exam PQ

Level 300 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria.
College of Science
Department of Physics
First Semester Examination

Academic Session: 2019/2020
Course Title: Analytical Mechanics I
Course Code: PHY313 (3 units)
Time Allowed: 3hrs.

27/02/2021

Instructions: Answer Question 1 and Any other Three Questions. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

Question 1

- A. Prove that the total angular momentum of a system of particles, referred to a given origin, can be expressed as the sum of two parts: one part being the angular momentum of all the particles relative to the centre-of-mass, and the other being the angular momentum of the total mass assumed to be at the centre-of-mass, relative to the origin.
- B. A system of particles consist of four particles of masses one quarter ($\frac{1}{4}$) unit each, with position vectors $\vec{r}_1 = 4t\hat{a}_x - 8\hat{a}_y - 4t^2\hat{a}_z$, $\vec{r}_2 = 12t\hat{a}_x + 4t\hat{a}_y$, $\vec{r}_3 = \vec{r}_1$, $\vec{r}_4 = \vec{r}_2$, where t is the time. Calculate the total angular momentum of the system.

(22 marks)

Question 2

- A. (i) State the Divergence theorem in Cartesian coordinates, and define all terms.
(ii) Given a force $\vec{F} = x^2\hat{a}_x + xy\hat{a}_y + yz\hat{a}_z$, in Cartesian coordinate. Verify the divergence theorem over a cube half ($1/2$) unit on each side. The cube is located at the first octant of the Cartesian system with one corner at the origin.
- B. State the curl of a vector field in Cartesian coordinates, and explain the physical meaning of the expression. (ii) Show that $\text{curl } \vec{B} = 0$, if $\vec{B} = (k/r)\hat{a}_r$, in spherical coordinates, where k is a constant.

(16 marks)

Question 3

- A. Consider a particle moving in a spherical coordinate system. Write down the coordinate transformation equations and obtain the position vector, $\vec{r}$ in spherical coordinate.
- B. Derive an expression for the kinetic energy and potential energy possess by the particle.
- C. Hence, obtain the Lagrangian formulations and derive the Larange's equations for the three generalized coordinates.

(16 marks)

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Question 4

A. Let (r, θ) represents the plane polar coordinates describing the position vector of a particle P. If $\hat{r}$ is a unit vector in the direction of the position vector $\vec{r}$, and $\hat{\theta}$ is a unit vector perpendicular to $\hat{r}$ and in the direction of increasing θ .

- Derive an expression for the velocity of the particle.
- Derive an expression for the acceleration of the particle.
- Show that the differential equation of the path obeyed by the particle of mass m , in terms of u and θ is $f(\pm) = -m(hu)^2 \left(u + \frac{d^2u}{d\theta^2} \right)$.

B. Assuming a particle is under the influence of a central force field $f(r) = \left(\frac{\alpha}{r^2} + \frac{\beta}{r^3} \right)$, where $\alpha, \beta > 0$. Find the orbit or path of the particle.

(16 marks)

Question 5

- Define central force motion and state three characteristics.
- State Kepler's laws of planetary motion, and prove the 3rd law.
- (i) Show that the magnitude of the velocity of a particle at any point on its path as it orbits elliptically the sun is $v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$.

Define the terms v, μ, a, r as it appears in C(i) above.

(16 marks)

Question 6

- Write down the point and vector transformations from (i) Cartesian to Spherical coordinates (ii) Spherical to Cartesian coordinates.
- (i) Given that vector $\vec{A} = A_r \hat{a}_r + A_\theta \hat{a}_\theta + A_z \hat{a}_z$, derive the transformation equation from Cylindrical to Cartesian coordinates in matrix form.

(16 marks)

$$\frac{1}{1} \propto \frac{1}{2} = \frac{1}{2}$$

$$\frac{1}{2}$$

Paper 6 of 7 — PHY 313 2017 2018

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Federal University of Petroleum Resources Effurun, Nigeria.
College of Science
Department of Physics
First Semester Examination

Academic Session: 2017/2018

Course Title: Analytical Mechanics I

Course Code: PHY313 (3 units)

Time Allowed: 3hrs.

29/05/2018

Instructions: Answer Question 1 and Any other Three Questions. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

Question 1

- A. Prove that the total angular momentum of a system of particles, referred to a given origin, can be expressed as the sum of two parts: one part being the angular momentum of all the particles relative to the centre-of-mass, and the other being the angular momentum of the total mass assumed to be at the centre-of-mass, relative to the origin.
- B. A system of particles consist of four particles of masses one quarter ($\frac{1}{4}$) unit each, with position vectors $\vec{r}_1 = 4t\hat{a}_x - 8\hat{a}_y - 4t^2\hat{a}_z$, $\vec{r}_2 = 12t\hat{a}_x + 4t\hat{a}_y$, $\vec{r}_3 = \vec{r}_1$, $\vec{r}_4 = \vec{r}_2$, where t is the time. Calculate the total angular momentum of the system.

(22 marks)

Question 2

- A. (i) State the Divergence theorem in Cartesian coordinates, and define all terms.
(ii) Given a force $\vec{F} = x^2\hat{a}_x + xy\hat{a}_y + yz\hat{a}_z$, in Cartesian coordinate. Verify the divergence theorem over a cube half ($1/2$) unit on each side. The cube is located at the first octant of the Cartesian system with one corner at the origin.
- B. State the curl of a vector field in Cartesian coordinates, and explain the physical meaning of the expression. (ii) Show that $\text{curl } \vec{B} = 0$, if $\vec{B} = (k/r)\hat{a}_r$, in spherical coordinates, where k is a constant.

(16 marks)

Question 3

- A. Consider a particle moving in a spherical coordinate system. Write down the coordinate transformation equations and obtain the position vector, $\vec{r}$ in spherical coordinate.
- B. Derive an expression for the kinetic energy and potential energy possess by the particle.
- C. Hence, obtain the Lagrangian formulations and derive the Lagrange's equations for the three generalized coordinates.

(16 marks)

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Question 4

A. Let (r, θ) represents the plane polar coordinates describing the position vector of a particle P. If $\hat{r}$ is a unit vector in the direction of the position vector $\vec{r}$, and $\hat{\theta}$ is a unit vector perpendicular to $\hat{r}$ and in the direction of increasing θ .

- Derive an expression for the velocity of the particle.
- Derive an expression for the acceleration of the particle.
- Show that the differential equation of the path obeyed by the particle of mass m , in terms of u and θ is $f\left(\frac{1}{r}\right) = -m(hu)^2 \left(u + \frac{d^2u}{d\theta^2} \right)$.

B. Assuming a particle is under the influence of a central force field $f(r) = \left(\frac{\alpha}{r^2} + \frac{\beta}{r^3} \right)$, where $\alpha, \beta > 0$. Find the orbit or path of the particle.

(16 marks)

Question 5

- Define central force motion and state three characteristics.
- State Kepler's laws of planetary motion, and prove the 3rd law.
- (i) Show that the magnitude of the velocity of a particle any point of its path as it orbits elliptically the sun is $v = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$.
(ii) Define the terms v, μ, a, r as it appears in C(i) above.

- Derive an expression for the velocity of the particle.
- Derive an expression for the acceleration of the particle.

(16 marks)

Question 6

- Write down the point and vector transformations from (i) Cartesian to Spherical coordinates (ii) Spherical to Cartesian coordinates.
- (i) Given that vector $\vec{A} = A_1\hat{a}_1 + A_2\hat{a}_2 + A_3\hat{a}_3$, derive the transformation equation from Cylindrical to Cartesian coordinates in matrix form.
(ii) Transform the vector field $\vec{A} = \hat{a}_r \sin \phi + \hat{a}_\phi \cos \phi + \hat{a}_z$, to Cartesian coordinates.

(16 marks)

Question 5

- Define central force motion and state three characteristics.
- State Kepler's laws of planetary motion, and prove the 3rd law.
- (i) Show that the magnitude of the velocity of a particle any point of its path as it orbits elliptically the sun is $v = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$.
(ii) Define the terms v, μ, a, r as it appears in C(i) above.

(16 marks)

Question 6

- Write down the point and vector transformations from (i) Cartesian to Spherical coordinates (ii) Spherical to Cartesian coordinates.
- (i) Given that vector $\vec{A} = A_1\hat{a}_1 + A_2\hat{a}_2 + A_3\hat{a}_3$, derive the transformation equation from Cylindrical to Cartesian coordinates in matrix form.
(ii) Transform the vector field $\vec{A} = \hat{a}_r \sin \phi + \hat{a}_\phi \cos \phi + \hat{a}_z$, to Cartesian coordinates.

(16 marks)

Paper 7 of 7 — PHY 313 2016 2017 ANALYTICAL MECHANICS 1

Level 300 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria.
College of Science
Department of Physics
First Semester Examination

Academic Session: 2016/2017
Course Title: Analytical Mechanics I
Course Code: PHY313 (3 units)
Time Allowed: 3hrs.

/2017

Instructions: Answer Question 1 and Any other Three Questions. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

Question 1

A. Prove that the total angular momentum of a system of particles, referred to a given origin, can be expressed as the sum of the angular momentum of all particles relative to the center of mass (C.M.), and the angular momentum of the total mass assumed to be the C.M. relative to the origin.

B. Three particles of masses $\frac{1}{2}$ unit each moves under the influence of a force field so that their position vectors relative to a fixed coordinate system are given respectively by $\vec{r}_1 = 3(t\hat{i} - t^2\hat{j} + t^3\hat{k})$, $\vec{r}_2 = 3(t^2\hat{i} + t^2\hat{j} + \hat{k})$ and $\vec{r}_3 = 3(t^2\hat{i} + t\hat{j})$, where t is the time. Calculate the total angular momentum of the system.

(22 marks)

Question 2

A. (i) Explain the terms conservative and non-conservative force fields and give two examples of each. (ii) Write down the appropriate expressions for the Nabla operator (∇) in Cartesian, Cylindrical and Spherical coordinates.

B. Determine which of the following forces are conservative

(i) $\vec{F} = ar^2 \cos(\phi)\hat{r} - ar^2 \sin(\phi)\hat{\phi} + 2ar^2 \cos(\phi)\hat{z}$

(ii) $\vec{F} = \{-2a \sin(\theta) \cos(\phi)\}\hat{r} - \{ar \cos(\theta) \cos(\phi)\}\hat{\theta} + \{\sin(\theta) \cos(\phi)\}\hat{\phi}$

(iii) $\vec{F} = \{3abyz^2 - 10bx^2y^2 + 2\}\hat{x} + \{7 + 3abxz - 5bx^2y\}\hat{y} + \{3 + 9abxyz^2\}\hat{z}$

C. (i) Obtain the potential function for the conservative force(s) in question 2(B) above.

(ii) Give the physical meanings of $\vec{F}$, F_r , F_θ , F_ϕ , $\vec{F}$, F_r , F_θ , F_ϕ as it appears in C(i) above.

(16 marks)

Question 3

A. Consider a particle moving in a spherical coordinate system. Write down the coordinate transformation equations and obtain the position vector, $\vec{r}$ in spherical coordinate.

B. Derive an expression for the kinetic energy and potential energy possess by the particle.

C. Hence, obtain the Lagrangian formulations and derive the Lagrange's equations for the three generalized coordinates.

(16 marks)

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Question 4

- A. Write down and define each term in the expression of the following:
 (i) Coriolis acceleration (ii) centripetal acceleration (iii) centrifugal acceleration (iv) linear acceleration (v) Coriolis force (vi) centripetal force (vii) centrifugal force.
- B. The angular momentum $\vec{L}$ of an XYZ coordinate system rotating about z-axis is given by $\vec{L} = \cos t \hat{i} + \sin t \hat{j}$ relative to a fixed coordinate XYZ coordinate system where t is the time. The origin of XYZ system has position vector $\vec{R} = t \hat{i} - \hat{j} + t^2 \hat{k}$ with respect to the XYZ system. Assuming that the position vector of the particle is given by $\vec{r} = (3t+1) \hat{i} - 2t \hat{j} + 5t \hat{k}$ relative to the moving system. At any time t , calculate (i) the apparent velocity (ii) true velocity (iii) apparent acceleration (iv) true acceleration (v) centripetal acceleration.

(16 marks)

Question 5

- A. Let (r, θ) represents the plane polar coordinates describing the position vector of a particle P. If $\hat{r}$ is a unit vector in the direction of the position vector $\vec{r}$, and $\hat{\theta}$ is a unit vector perpendicular to $\hat{r}$ and in the direction of increasing θ .
 (i) Derive an expression for the velocity of the particle.
 (ii) Derive an expression for the acceleration of the particle.
 (iii) Show that the differential equation of the path obeyed by the particle of mass m , in terms of r and θ is $f(\theta) = -m(m)^2 \left(u + \frac{d^2 u}{d\theta^2} \right)$.

- B. Assuming a particle is under the influence of a central force field $f(r) = \left(\frac{\alpha}{r^2} + \frac{\beta}{r^3} \right)$, where

$\alpha, \beta > 0$. Find the orbit or path of the particle.

(16 marks)

Question 6

- A. Define central force motion and state three characteristics.
 B. State Kepler's laws of planetary motion, and prove the 3rd law.
 C. (i) Show that the magnitude of the velocity of a particle any point of its path as it orbits elliptically the sun is $v = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$.
 (ii) Define the terms v , μ , a , r as it appears in C(i) above.

(16 marks)